

# Xining Li

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Canadian Citizen · TN-visa eligible · on-site / hybrid

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## PROFILE

Software engineer specializing in **applied ML and ML infrastructure** — **speech/audio, reinforcement learning, and generative models** — with six years shipping production systems at worldwide scale. At **Microsoft** I was the **on-call Designated Responsible Individual** for **Microsoft 365 Copilot Business Chat**, an LLM-powered service in worldwide production: I **triaged, debugged, and resolved Sev-2+ incidents**, mitigated latency regressions, and built the **telemetry, monitoring, and evaluation** pipelines that kept the service honest. Alongside production work I write the product/system code (**Python, Go, TypeScript**) and the **evaluation harnesses** (blind gold standards, leakage-free splits, graded agent evaluation) for my own ML systems — with the testability discipline (mutation testing, contract tests, mock frameworks) a code-review-driven team expects. Two master's degrees (**M.A. Statistics, MEL — Dependable Software Systems**); first-author paper at **ALIFE 2026**.

## EXPERIENCE

### Microsoft — Software Engineer

Oct 2022 – Sept 2026

Vancouver, BC · Microsoft 365 Copilot Business Chat — LLM + speech/audio services at worldwide scale · Python, C#, TypeScript, Kusto/KQL, Azure

- **Production ownership & incident response:** **primary on-call Designated Responsible Individual** for **Microsoft 365 Copilot Business Chat** in worldwide production — **triaged, debugged, tracked, and resolved Sev-2+ incidents** by analyzing root cause and impact on service quality; mitigated **latency regressions** and deployment blockers; authored post-incident reviews and on-call handoff runbooks — end-to-end production **debugging** of a worldwide LLM-powered service.
- **Speech/audio:** contributed to the **real-time WebRTC voice path** for human-to-agent calls (media pipeline, latency and reliability tuning, client-server integration) and to the **staged rollout configuration** for real-time communication features across Teams, Outlook, and Edge; applied a **TurnSense-style sound-classification technology** routing **non-semantic sounds** (Windows notification chimes, Teams alert tones) into an **Invalid** category to be **silently ignored** — blocking false triggers at the source.
- **Reinforcement learning + model evaluation:** designed and built an **end-to-end evaluation environment** for **Microsoft 365 Copilot Business Chat** — a multimodal (voice + UI) product — a **model-agnostic contract layer** (audio injection, UI observation/action, session recording) driving a **computer-use agent** that autonomously exercises both the audio and UI surfaces in a single **automated test**. Defined the **evaluation metrics and reward criteria**, then progressively **fine-tuned the agent** through a **human-in-the-loop RLHF/DPO cycle**, reporting results to stakeholders.
- **Telemetry, monitoring & data processing:** built Business Chat's diagnostic / error-logging and telemetry pipelines — event tracking, privacy review, log-volume optimization, queried in **Kusto/KQL** (Azure Data Explorer) — and authored the operational dashboards (**Jarvis**) for latency, availability, QoS, resolved-ratio, failure-rate, and scenario categorization. Production **service monitoring and data processing** for an LLM-powered service.

### Kidoz — Backend Engineer

Jan 2022 – Oct 2022

Vancouver, BC · Go, Python, PHP/Laravel, MySQL (deep production use), Redis, Docker

- **Product/system code & distributed systems in Go:** implemented **distributed-coordination primitives** — a Redis-backed **spin lock** and a **leader-election** system for service failover — plus an ad-bidding metadata framework (Google Play / App Store) with a full mock-based test suite and a video upload/transcoding service. **Data structures and algorithms** in production backend code.
- **ML in production:** built a **content-understanding classification system** for image/video assets with **confidence-scored human-in-the-loop (HITL) review** to filter unsafe / age-inappropriate material; designed a lightweight **explicit-signal-based ranking pipeline** (**offline evaluation in Python, online serving in Go**) prioritizing diversity and safety over engagement, with strict **data minimization** in real-time feature extraction — **fully compliant with GDPR (specifically GDPR-K) and COPPA**. Model deployment and data processing on a live serving path.
- **Testability & correctness:** campaign-service validation framework with transactional **MySQL** tests in Docker; day-to-day **MySQL** on the ad-serving path (schema design, index and query tuning at scale). Code-review-grade test discipline.

### Apple (contract) — Big Data DevOps Engineer

Aug 2019 – Jan 2021

San Jose, CA · Apple AD Platform — YARN, HDFS, Oozie · map-reduce / streaming / Scala indexing pipelines · Elasticsearch, Solr · Python, SQL/Hive · Splunk, AWS EC2

- **Data infrastructure at scale:** operated, tested, and **debugged large-scale Hadoop data pipelines** (YARN, HDFS, Oozie) and the **offline indexing pipelines** (map-reduce, streaming, Scala) feeding **Elasticsearch / Solr** across review and staging environments — the **data-processing** layer at the scale of billions of records.
- **Pager-duty on-call & deployment:** carried **pager-duty on-call** — **debugged end-to-end Oozie workflows** from **Splunk** log inspection and, on confirmed failure, **shut down the affected AWS EC2 containers**; shipped pipeline changes through **blue-green** deployments.
- **Testability & CI:** designed an end-to-end **mock YARN/HDFS/Oozie framework** enabling full pipeline testing **without live-cluster dependencies**, plus **mock-data testing solutions** for the offline indexing pipelines and search engines with **end-to-end tests against online servers** across **asynchronous offline/online processes**; introduced **branch-coverage** and **mutation-testing** frameworks — quantitative CI gates on correctness, not just “tests pass”.

### HG Insights (formerly HG Data) — Data Scientist

Oct 2018 – Mar 2019

Santa Barbara, CA · Python, SQL

- **ML specialization (statistical modeling):** designed the company-product **purchase-propensity ranking model** — windowed selective sampling into per-product design matrices, caching and parallel computation. **+20% sales leads** in client **A/B testing** (patent pending, App. #505502946); model evaluation by randomized experiment, plus the data processing underneath.

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## SELECTED ML / INFRASTRUCTURE PROJECTS

**ALIFE 2026 first-author paper — Dreaming Itself: Diffusion Models as the Structural Substrate for Self-Awareness in Humanoid Robots**

2026

ISAL / MIT Press · [direct.mit.edu/isal/proceedings/isal2026/38/125/138158](https://direct.mit.edu/isal/proceedings/isal2026/38/125/138158) · companion code: [github.com/xiningli/dreaming-together](https://github.com/xiningli/dreaming-together)

**CASA — speech/audio + multimodal ML and an evaluation harness**

2026

Private research collaboration · PyTorch, ArcFace, ECAPA, von Mises-Fisher · single 16 GB GPU

- **Speech/audio + multimodal ML**: attributed every subtitle line to its on-screen speaker by fusing **ECAPA voice embeddings** (audio), **ArcFace** face recognition (vision), and active-speaker detection under a **von Mises–Fisher posterior**; Viterbi decode → **Platt-calibrated** confidence triaging human review — cross-modal **speech/audio ML**.
- **ML evaluation infrastructure**: longitudinal corpus with per-utterance metadata, **week-based leakage-free splits machine-verified at zero leakage** (a CI gate), and a **gold-standard protocol** as a first-class artifact — blind labeling, append-only per-edit audit logs, gold frozen on merge. Model-evaluation rigor treated as infrastructure.

**WebTask Arena — RL environments, model serving & graded evaluation** 2026  
*webtask-arena · FastAPI, Docker, Playwright, Hugging Face Spaces*

- **RL environments + model deployment**: built **deterministic, seed-reproducible, server-graded environments** with **named-subgoal reward functions** for computer-use agents, plus a **contract-test suite** for determinism and reward correctness. **FastAPI** service, **Docker**, **deployed live on Hugging Face Spaces** — the RL → model-serving → cloud-deployment loop.
- **Model evaluation & serving**: graded evaluation of **UI-TARS-1.5-7B** (4-bit, single 16 GB GPU) over a Playwright screenshot → action loop with **VLM serving (vLLM, Qwen2.5-VL)**; diagnosed failure modes from recorded trajectories (**uitars-webtask-eval**) and fed them back into reward and environment design.

**TECHNICAL SKILLS**

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>                          | <b>Python</b> (6 yrs production: PyTorch, scikit-learn, pandas, FastAPI), Go, TypeScript / JavaScript, PHP, <b>SQL</b> (MySQL — deep production: schema design, index & query tuning), Kusto/KQL                                                                                                                                                                                                                                                                                                                                                                      |
| <b>ML infrastructure</b>                  | <b>model deployment</b> (Docker, FastAPI/REST, Hugging Face Spaces, <b>vLLM</b> serving, microservices), <b>model evaluation</b> (blind gold standards, leakage-free splits as CI gates, graded agent eval, Platt calibration), <b>model optimization</b> (4-bit quantization, latency tuning), <b>data processing</b> (Hadoop YARN/HDFS/Oozie, map-reduce / streaming / Scala indexing pipelines, Elasticsearch / Solr, Kusto/KQL telemetry), <b>production debugging</b> (on-call DRI for a worldwide LLM-powered service, Sev-2+ incidents, post-incident reviews) |
| <b>Specialized ML areas</b>               | <b>speech/audio</b> (production, contributed: WebRTC real-time voice path, media pipeline, staged RTC rollouts, TurnSense-style non-semantic-sound classification for false-trigger suppression; research ML: ECAPA voice embeddings, von Mises–Fisher speaker posterior, CosyVoice2), <b>reinforcement learning</b> (reward / graded environments, computer-use agents, progressive fine-tuning via human-in-the-loop RLHF/DPO), <b>generative AI</b> (diffusion models — ALIFE 2026 paper), recommendation / ranking, multi-modal content understanding             |
| <b>Deep-learning frameworks</b>           | <b>PyTorch</b> , scikit-learn, <b>vLLM</b> (VLM serving), Qwen2.5-VL, UI-TARS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Software engineering &amp; systems</b> | Product/system code, <b>data structures &amp; algorithms</b> (Redis spin lock, leader election), REST / microservices, distributed systems, <b>mutation-testing + branch-coverage</b> CI gates, contract-test suites, mock frameworks, code-review / testability discipline                                                                                                                                                                                                                                                                                           |
| <b>Cloud</b>                              | <b>Azure</b> (Microsoft production worldwide), <b>AWS EC2</b> (Apple), Docker, Hugging Face Spaces                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Statistics &amp; quantitative</b>      | <b>M.A. Statistics</b> ; Bayesian inference (von Mises–Fisher, forward–backward, Viterbi, Platt calibration), <b>A/B testing</b> in production (+20%), robust estimation (Theil–Sen), causal-necessity ablation, propensity modeling                                                                                                                                                                                                                                                                                                                                  |

**EDUCATION**

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|----------------------------------------------------------------------------|------|
| <b>University of British Columbia</b>                                      | 2022 |
| <i>Master of Engineering Leadership (MEL), Dependable Software Systems</i> |      |
| <b>University of California, Santa Barbara</b>                             | 2019 |
| <i>Master of Arts in Statistics</i>                                        |      |
| <b>University of California, Santa Barbara</b>                             | 2017 |
| <i>Bachelor of Science in Financial Mathematics and Statistics</i>         |      |